

HealthShare: Improving healthcare efficiency and performance with Google Cloud

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With Google Cloud, HealthShare is reducing latency, improving performance and furthering its mission of empowering patients and providers to make better-informed healthcare decisions. HealthShare enables doctors to provide quality care.



About HealthShare

HealthShare's innovative digital platform helps patients and doctors make better-informed health decisions. By breaking down the information silos that exist in healthcare, HealthShare ensures the right information is provided to the right doctor or the right patient at the right time.

Industries: Healthcare

Google Cloud results

- Scales to support demand peaks of 6,000 requests per minute
- Minimizes downtime notifications through improved reliability
- Ensures compliance with United States legislation governing healthcare information

Halvin
to 50
millise
with G
Cloud

Location: Australia

Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine/>)

BigQuery (<https://cloud.google.com/bigquery/>)

Cloud SQL (<https://cloud.google.com/sql/>)

Cloud Load Balancing

(<https://cloud.google.com/load-balancing/>)

Google Cloud (<https://cloud.google.com/>)

Founded in 2010, HealthShare is a health technology company that connects and engages all health stakeholders (including patients, doctors, and healthcare organizations) to deliver optimal patient health outcomes. “Our mission is to help patients and doctors make better-informed health decisions through the use of innovative technology,” says Rami Weiss, co-founder of HealthShare.

Over the last decade, the organization has created a digital platform to improve healthcare delivery. “We are trying to tackle the silos of information that exist in the health ecosystem that mean the right information is not made available to the right people at the right time, so they can make the best possible healthcare decisions,” says Weiss.

“That situation causes waste and inefficiencies and means patients are not getting the best possible treatment and doctors don’t have all the critical information at their fingertips to make the best treatment decisions,” he adds. “It also means the doctor-patient interaction isn’t optimized for the limited time they have together”.

“By streamlining information flows, we can deliver better patient outcomes, improve doctor satisfaction, and support patients to take more control over their health.”

One of HealthShare’s flagship products is **BetterConsult** (<https://www.betterconsult.com/>), a clinical digital questionnaire patients fill out before a doctor’s appointment. This questionnaire captures information such as why a patient is seeing a doctor, what medications that patient is already taking, and the patient’s medical and social history. “We capture the patient’s current state of health and their needs in the form of a concise doctor’s note,” says Weiss. “This ensures the doctor can spend more time on satisfying the patient’s needs, while also spending less time siphoning through the patient’s history in the EMR.”

“The algorithms we’ve built are broad and complex in scope, covering symptoms, conditions, prescriptions, referrals, and other reasons that patients present to their doctor.”

Google Cloud powers all HealthShare products, including BetterConsult, which is currently available in Australia and the United States, and the Australia-only [HealthShare.com.au](https://healthshare.com.au/) (<https://healthshare.com.au/>) website. “That website receives over 800,000 monthly unique visitors and is primarily about giving consumers the information to find the right health professional for their needs,” says Weiss. “For example, if a consumer needs an orthopaedic surgeon who specializes in knees and operates from a certain hospital—and has ‘no gap’ coverage with that consumer’s health insurer—we have those disparate pieces of information in a centralized, easily searchable specialist directory that we can make available to patients and health professionals.”

A dynamic technology strategy

To accommodate rapid changes in its products and market, HealthShare has adopted a dynamic, near-term-focused technology strategy. “Our strategy is based on listening, learning, and iterating towards our goals,” says Lachlan Cannon, Chief Technology Officer at HealthShare. “We also try to minimize the time we spend on underlying systems and infrastructure, so we can focus on our stakeholders’ needs.”

The business started operations running virtual machines in a hosted data center environment. “We

initially ran that infrastructure the same way we would operate on-premises or owned hardware,” says Cannon. “We then moved to scripting, making repeatable processes part of our code base and creating virtual machines on the fly.” However, HealthShare began to experience reliability issues and, by the time Google Cloud launched its Sydney region in July 2017, was ready to move to a new provider.

“We experienced reliability problems at least once a quarter and sometimes as frequently as multiple times a month,” says Cannon. “We sometimes saw our website grind to a halt during high-traffic periods when we ran on machines with other tenants that were also busy.”

With BetterConsult in development at that stage, HealthShare was primarily concerned about the reliability of HealthShare.com.au. The website features tools that integrate with general practitioners’ desktop clinical software to provide easily searchable directory and referral services. “If doctors see errors in the software or our tools aren’t available, they call our partners’ support services and begin to lose faith in our service,” says Weiss.

Reliability would also prove critical to the now fully operational BetterConsult. “If we send a text message to patients and our website fails to start up when they click through to fill out the questionnaire, there’s a good chance they’re not going to click through again,” says Weiss.

“So for us to deliver value, doctors need to have a consistent, reliable process in place when they see patients.”

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another
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—*Lachlan Cannon, Chief
Technology Officer,
HealthShare*

Network quality key to decision-making

The business evaluated competing multinational cloud services and elected to move to Google Cloud. HealthShare based its decision on the quality of Google Cloud’s network and the skills of its network engineers, its plan to move from virtual machines to containerization through Kubernetes, and the fact the Google Cloud team made contact and invited the organization to events to help educate its team about the platform.

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With HealthShare targeting the United States market with BetterConsult, Google Cloud also enabled the organization to comply effectively with the United States Health Insurance Portability and Accountability Act—a federal law that establishes

data privacy and security requirements for organizations responsible for protecting personal-protected health information.

The business then spent about three months porting its products, services, and data across to Google Cloud regions in Sydney and the United States. “The Google Cloud team reviewed our proposed architecture and answered all our questions during the migration process—helping ensure a smooth project,” says Cannon.

“Most of our
healthcare
clients want to
see reports into
services such as
promoted
listings, and it’s
very important
for us to
demonstrate
they are
receiving sound
returns on their
investments.
Good quality

analytics—
through
BigQuery— are
key to this.”

—Rami Weiss, co-founder,
HealthShare

Google Kubernetes Engine at heart of infrastructure

The HealthShare Google Cloud environment now comprises [Kubernetes](#) (/kubernetes) clusters—managed through [Google Kubernetes Engine](#) (/kubernetes-engine)—for its test and production environments. “We run our app servers, web server, task routers, web sockets, and queues internally, with [Cloud SQL](#) (/sql) for PostgreSQL running separately for our databases,” says Cannon. “[Cloud Load Balancing](#) (/load-balancing) acts as an entry point to this environment.” The HealthShare.com.au website and the Australian version of BetterConsult run in Sydney, with a second version of BetterConsult running in the United States in a similarly designed architecture.

The business has also turned to a [BigQuery](#) (/bigquery) analytics data warehouse for data analysis and reporting. “Most of our healthcare clients want to see reports into services such as promoted listings, and it’s very important for us to demonstrate they are receiving sound returns on

their investments,” says Weiss. “Good quality analytics are key to this.”

With Cloud SQL,
we can take
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high availability
option to
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and reduce our
management
requirements.”

—*Lachlan Cannon, Chief
Technology Officer,
HealthShare*

Latency halved

With Google Cloud, HealthShare has reduced the latency for average requests from about 100 milliseconds on the infrastructure run in the hosted datacenter to about 50 milliseconds—improving the performance and experience of its products for users.

While HealthShare's overall infrastructure costs are about the same as when the organization operated in the hosted facility, the organization has been able to support increased usage and improve its reliability. "For example, we hadn't replicated our database—with Cloud SQL, we can take advantage of a high availability option to minimize risk and reduce our management requirements," says Cannon. The infrastructure scales seamlessly to manage demand peaks for HealthShare.com.au that start about 8 AM each morning as doctors log into the website and reach about 6,000 requests per minute before falling to about 4,000 requests per minute throughout the day.

Rapid pandemic response

HealthShare's agility enabled the business to quickly add coronavirus pre-screening into the questionnaire workflow of BetterConsult. "Our team worked quickly to roll out a number of COVID-related questions based on Department of Health guidelines around patient prescreening before they arrive at medical practices and enable doctors to shift to telephone or telehealth consults," says Weiss. "Off the back of that campaign, we experienced massive increases in utilization, and growth every week has reached record highs. More importantly, we have heard from lots of practices that our prescreening has helped keep their doctors, practice staff, and patients safe and avoid unnecessary infections."

A bright future

Weiss describes the future for the business with Google Cloud as “very, very bright.”

“We have excellent products that we can continue building out the functionality and workflows for. If you look at our mission since we started the company—providing information to empower patients to make better-informed health decisions and empower doctors to provide the best quality care to patients—there’s huge scope beyond what we’ve built to date,” he adds. “So we will continue to execute and iterate on existing products, hear from customers and stakeholders about additional features and offerings we can plug into those platforms, and continue to expand our footprint internationally.”